

Math 421

Friday, November 21

Announcements

- Homework 9 due **today**
- Next week's drop-in hours:
 - **Tuesday** 9-11 am VV B205
 - Wednesday 12-1 pm VV 311
- Check for any final exam conflicts & email me

Chapter 3: Continuity

Section 19: Uniform Continuity

Uniform Continuity

Definition. Let f be a real-valued function defined on a set $S \subseteq \mathbb{R}$. Then f is **uniformly continuous** on S if:

for all $\epsilon > 0$ there exists $\delta > 0$ such that
 $x, y \in S$ and $|x - y| < \delta$ imply $|f(x) - f(y)| < \epsilon$.

Q: If f is uniformly continuous on its domain, is f continuous on its domain?

Uniform Continuity Proof Recipe

To show that $f: S \rightarrow \mathbb{R}$ is uniformly continuous on S :

- I. “Fix $\epsilon > 0$.”
- II. “Let $\delta = (\textit{something})$.” [this requires scratch work]
- III. “Assume _____ and _____.”
- IV. Show that _____.
- V. “Since $\epsilon > 0$ was arbitrary, f is uniformly continuous on S .”

Example

Let $S = \{x \in \mathbb{R} : -4 \leq x \leq 4\}$, show that $g(x) = x^2$ is uniformly continuous on S .

Activity – Find δ

1. Show that any constant function is uniformly continuous on $\mathbb{R}$.
2. Let $S = \{x \in \mathbb{R} : -N \leq x \leq N\}$, show that $g(x) = x^2$ is uniformly continuous on S .

Q: Do you think $g(x) = x^2$ is uniformly continuous on $\mathbb{R}$?

Continuity and Uniform Continuity

Fact. If f is uniformly continuous on its domain, then it is continuous on its domain.

Theorem 19.2. If f is a continuous function on $[a, b]$, then f is uniformly continuous on $[a, b]$.

Chapter 5: Differentiation

Section 28: Basic Properties of the Derivative

Differentiable

Definition. Let f be a real-valued function defined on an open interval containing a point a . We say f is **differentiable at a** , or f has a derivative at a , if the limit

exists and is finite.

Example

Compute f' if $f(x) = x^2$.

Activity

Compute f' if $f(x) = mx + b$, where m, b are constants.

Activity – Divide & Conquer

Let $f(x) = |x|$. Compute f' by dividing into 3 cases.

Case 1: $a > 0$

Case 2: $a < 0$

Case 3: $a = 0$

Differentiable Implies Continuous

Theorem 28.2. If f is differentiable at a , then f is continuous at a .

Q1: Is the converse of this statement true?

Q2: What might a function have to look like for it to be *continuous everywhere* but *differentiable nowhere*? It might be helpful to first think of the following things:

- What might it look like for a function to be continuous but not differentiable at a finite number of points?
- What might it look like for a function to be continuous but not differentiable at every integer?